

Md Ajwad Akil

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RESEARCH INTEREST

Software and Network Security, Natural Language Processing(NLP), Large Language Models(LLM) and it's security, Machine Learning for security, Retrieval Augmented Generations(RAG), Adversarial Machine Learning

EDUCATION

Purdue University

PhD in Computer Science

Indiana, USA

Aug 2023 – Present

- **Supervisor:** Dr. Elisa Bertino
- **CGPA:** 3.8/4.00
- **Courses Taken:** Deep Learning, Natural Language Processing, Algorithms Design and Analysis

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

Feb 2017 – May 2022

- **Thesis:** Readable Bangla Text and Paraphrase Generation
- **Supervisor:** Dr. Rifat Shahriyar
- **CGPA:** 3.82/4.00

RESEARCH EXPERIENCE

Graduate Student

Purdue University

April 2024 - Present

Indiana, USA

- Cyber Space Security Lab (Cyber2Slab, PI: Dr. Elisa Bertino)

Undergraduate Research Assistant

Bangladesh University of Engineering and Technology (BUET)

March 2021 - Oct 2022

Dhaka, Bangladesh

- BUET CSE NLP Group (PI: Dr. Rifat Shahriyar)

PUBLICATIONS

- Gourab Saha, Shashata Sawmya, Arpita Saha, **Md Ajwad Akil**, Sadia Tasnim, Md Saifur Rahman, M Sohel Rahman - **PRIEST: predicting viral mutations with immune escape capability of SARS-CoV-2 using temporal evolutionary information** - *Briefings in Bioinformatics*
- **Md Ajwad Akil**, Najrin Sultana, Abhik Bhattacharjee, Rifat Shahriyar - **BanglaParaphrase: A High-Quality Bangla Paraphrase Dataset** - *IJCNLP-AAACL 2022*
- Ishrat Jahan Eliza, Md Hasibul Husain Hisham, Mohammad Nuwaisir Rahman, **Ajwad Akil**, Abir Mohammad Turza, Fahim Morshed, Nazmus Sakib, Sriram Chellappan, A. B. M. Alim Al Islam - **CORONOSIS: Corona Prognosis via a Global Lens to Enable Efficient Policy-making Both at Global and Local Levels** - *ACM COMPASS - 2022*

RESEARCH PROJECTS

Malware Analysis with Large Language Models

Supervisor: Dr. Elisa Bertino, Dr. Imtiaz Karim (Purdue University)

April 2024 - Present

- The objective of the project is to leverage open-source LLM for malware analysis and generation.
- Currently we are investigating whether we can leverage off-the-shelf LLM by incorporating zero-shot prompting methods to analyze and generate malware variants. We are conducting extensive experiments with different malware representations(code, binaries, etc) by incorporating LLM into an extensive pipeline for malware variant generation.

PRIEST

Supervisor: Dr. M. Sohel Rahman (BUET)

Feb 2022 - Aug 2023

- We developed an interpretable and explainable Deep Learning model to predict mutations in the SARS-CoV-2 virus using its evolutionary information where we trained a mutation predictor by modifying Tempel for our protein sequences using a base Transformer beating the baseline predictors. We used it to generate mutations and tested the quality through various methods. We also tested a variant classifier on these mutated sequences and validated our approach through several methods.
- I worked on embedding generation and Classification model training and hyperparameter optimization. I also worked on the mutation prediction pipeline modification and the validation of our approach by using clustering and immune escape methods.

BanglaParaphrase: A High-Quality Bangla Paraphrase Dataset

March 2021 - Oct 2022

Supervisor: Dr. Rifat Shahriyar (BUET)

- We curated an all-synthetic Bengali paraphrase dataset and a baseline paraphrase-generating model and developed a novel filtering pipeline that ensured diversity and semantics within the dataset and experimented with POS tagging along with MLM to diversify our existing dataset.
- I was involved in filtering and evaluation metric selection by rigorous literature review, developed some parts of the filtering pipeline, modified the BERTScore codebase to use BanglaBERT, fine-tuned the models with our dataset, and worked on the write-up of the project.

CORONOSIS

May 2020 - Jan 2021

Supervisor: Dr. A. B. M. Alim Al Islam (BUET)

- We developed an integrated dashboard for the COVID-19 pandemic where we performed several predictions (next-day predictions, when would we keep coronavirus under control, country-wise affected and dead, etc.) using epidemiology models. These analyses were demonstrated through interactive charts, graphs, and maps and also explored the correlations between disease spread and diverse socio-economic and environmental factors.
- I conducted data collection and curation, developed of data processing pipeline, worked on the back-end part of the dashboard as well and worked on the write-up of the project.

ACADEMIC SERVICES

Sub-Reviewer

- **Conference:** Network and Distributed System Security (NDSS) 2025, Annual Computer Security Application (ACSAC) 2024, International Symposium on Research in Attacks, Intrusions, and Defenses (RAID) 2024
- **Journal:** IEEE ACM Transactions on Networking

Artifact Evaluation Committee (AEC) Member

- Annual Computer Security Application (ACSAC) 2024

PROFESSIONAL EXPERIENCE

Lecturer

Dhaka, Bangladesh

Employer: United International University

July 2022 - July 2023

- I worked as a lecturer in the Department of Computer Science and Engineering where I instructed various courses such as Advanced OOP, Structured Programming, Data Structures, and Algorithms.

AI Engineering Intern

Dhaka, Bangladesh

Employer: Intelligent Machines

July 2021 - June 2022

- I Worked extensively on Deep Learning projects such as noise-robust audio sound event detection from raw audio streams, OCR based projects on real-world Datasets(identity cards, financial statements, etc.) as well as deployment as part of POC.

ACADEMIC PROJECTS

A comparison Between Knowledge Distillation and Few-Shot Learning

2024

- Performed a comparative analysis between Representation Distillation and Few-Shot Learning on MedMNIST V2 Dataset

Prompt Enhanced Medical Question Summarization

2024

- We investigated how we can use NER, co-occurrence, and instruction fine-tuning to summarize medical questions.

Image Similarity Search with Autoencoders

2022

- This project retrieves images using embeddings learned with the help of autoencoders and variational autoencoders.

CNN from scratch 2022

- Implemented a CNN from scratch with NumPy with support for a variable number of layers, dimensions, and batching.

Hidden Markov Model with Viterbi and Baum Welch Algorithm 2021

- Hidden Markov Model implemented with Viterbi algorithm and Baum-Welch Algorithm.

Classical AI and Machine Learning Projects 2020

- Several classical AI and ML-based projects such as an Adversarial search-based game solver, CSP-based Latin square solver and implementing logistic regressions and AdaBoost algorithms from scratch.

AWARDS

- Dean's List Scholarship: Bangladesh University of Engineering and Technology (BUET)
- Board Merit Scholarship: Government of Bangladesh (Year: 2014, 2016)

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript

Database: PostgreSQL, MongoDB, Elastic Search

Frameworks: Tensorflow, PyTorch, PyTorch Geometric, Node.js, React.js, React-native, Javafx, OpenGL

Tools/Software: Git, Microsoft Office Suit, Figma, MATLAB, Latex, Adobe XD, Cisco Packet Tracer, Wireshark

REFERENCE

- **Dr. Elisa Bertino**
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Purdue University
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- **Dr. Rifat Shahriyar**
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- **Dr. Imtiaz Karim**
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